

AADK 11: Task 1 — Data-Source Design & List Structure Planning

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Course/Unit: Android Development with Kotlin – Session 11

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1. Theme Selection

Theme: Travel Destinations

This list will display famous travel destinations around the world. Each item will include an image, title, location, description, and a favourite flag.

2. Data Model Design

Kotlin Data Class

```
data class TravelDestination(  
    val title: String,  
    val country: String,  
    val imageResId: Int,  
    val description: String,  
    val isFavourite: Boolean  
)
```

Field Explanation

Field	Type	Purpose
title	String	Name of the destination
country	String	Country where the destination is located
imageResId	Int	Reference to drawable image resource

Field	Type	Purpose
description	String	Short description of the place
isFavourite	Boolean	Indicates if the destination is marked as favourite

3. Sample Dataset (10 Items)

val destinations = listOf(

 TravelDestination("Eiffel Tower", "France", R.drawable.destination1, "Iconic landmark in Paris.", false),

 TravelDestination("Great Wall", "China", R.drawable.destination2, "Ancient world wonder stretching across China.", false),

 TravelDestination("Taj Mahal", "India", R.drawable.destination3, "Beautiful marble mausoleum in Agra.", true),

 TravelDestination("Grand Canyon", "USA", R.drawable.destination4, "Massive canyon carved by the Colorado River.", false),

 TravelDestination("Colosseum", "Italy", R.drawable.destination5, "Historic Roman amphitheatre.", true),

 TravelDestination("Mount Fuji", "Japan", R.drawable.destination6, "Japan's famous snow-capped mountain.", false),

 TravelDestination("Sydney Opera House", "Australia", R.drawable.destination7, "Iconic performing arts centre.", false),

 TravelDestination("Christ the Redeemer", "Brazil", R.drawable.destination8, "Large statue overlooking Rio.", false),

 TravelDestination("Pyramids of Giza", "Egypt", R.drawable.destination9, "Ancient pyramids built thousands of years ago.", true),

 TravelDestination("Santorini", "Greece", R.drawable.destination10, "Famous island with white buildings and blue domes.", false)

)

4. UI Layout Plan (Wireframe)

Item Layout Concept



Compose Components to Use

Component Purpose

LazyColumn Displays scrollable list

items{} Iterates through dataset

Card Creates card style layout

Row Align image and text horizontally

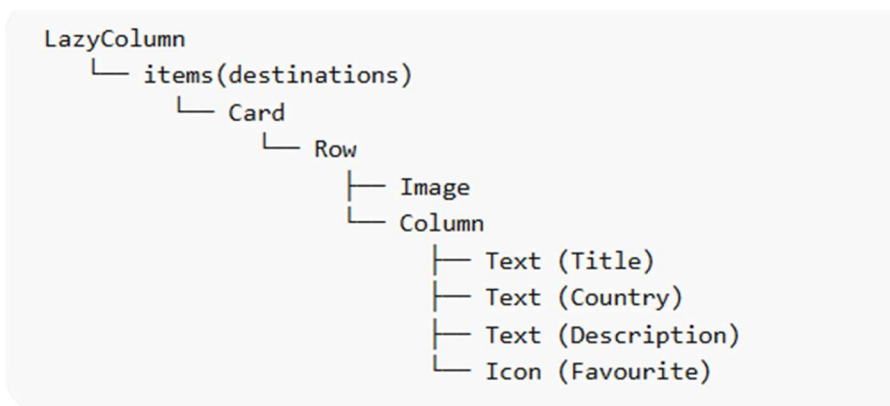
Column Organize text vertically

Image Displays destination image

Text Displays title, country, description

Icon Shows favourite status

Layout Structure (Concept)



Spacing & Padding

- Card Padding: **8dp**
- Image Size: **80dp**
- Text Spacing: **4dp**
- Row Padding: **12dp**

Favourite Indicator

If isFavourite = true:

- Show **filled heart icon**
- Icon color = **Red**

If false:

- Show **outlined heart icon**
-

5. Navigation / Interaction Plan (Bonus)

When the user taps on a destination item, the app will navigate to a **detail screen** showing a larger image and full description of the destination.

If the user taps the **heart icon**, the isFavourite state will toggle between true and false, updating the icon appearance.

6. Reflection

Designing the data model and UI layout before coding helps organize the structure of the application. It ensures that the data fields match the UI elements needed to display them. Planning also prevents mistakes such as missing data fields, poorly structured layouts, or inefficient state handling. By thinking through the structure early, developers can write cleaner, more maintainable code and reduce debugging time later.